a. this period (teenage) is spread almost over a decade,
b. it is characterized by rapid increase in height and weight, hormonal changes, sexual maturation and wide swings in emotion,
c. development of critical bone mass is essential during this period as this forms the ground for maintaining mineral integrity of the bone in later life,
d. the pattern and proportion of various body components like body water, muscle mass, bone and fat increase during the entire childhood and adolescence to reach adult values by about 18 years.

Adolescent girls are at greater physiological stress than boys because of menstruation. Their nutritional needs are of particular importance as they have to prepare for motherhood.

Nutritional Needs of Adolescents

Good nutrition is critical during the teenage years to ensure healthy growth and development. A healthy diet must meet the changing nutritional needs of a growing teenager.

Energy

Calorie needs is influenced by activity level, basal metabolic rate, increased requirements to support pubertal growth, development and energy expenditure. The energy requirements for boys are more than that of girls.

Protein

Protein needs of adolescents are influenced by the amount of protein required for maintenance of existing lean body mass and accrual of additional lean body mass during the adolescent growth spurt. When protein intakes are consistently inadequate, reductions in linear growth, delays in sexual maturation and reduced accumulation of lean body mass may be seen.

Fat and Essential Fatty Acids

The human body requires dietary fat and essential fatty acids for normal growth and development. The intakes of total fat and saturated fat should not exceed RDA.

Calcium

Calcium needs during adolescence are greater than they are in either childhood or adulthood because of the dramatic increase in skeletal growth. Milk provides the greatest amount of calcium in the diets of adolescents.Ragi, green leafy vegetables, milk and milk products are excellent sources of calcium.

Iron

Iron is vital for transporting oxygen in the bloodstream and for preventing anaemia. For both male and female adolescents, the need for iron increases with rapid growth and the expansion of blood volume and muscle mass. The onset of menstruation imposes additional iron needs for girls.

Zinc

Zinc is important in adolescence because of its role in growth and sexual maturation. Males who are zinc deficient experience growth failure and delayed sexual development.

Vitamins

Vitamin A is important for normal vision and plays a vital role in reproduction, growth, and immune function. The most obvious symptom of inadequate vitamin A

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